Bots and the Ballot Box: How Automated Accounts Shaped Narratives and Networks in the 2024 US Election

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Abstract

Social bots are increasingly deployed to sway public opinion during high-stakes events. We analyze 42 million election-related tweets from X (formerly Twitter), label 4.3 million accounts with BotMoE, and compare the temporal behavior, topical focus, and network position of bots versus human users. Bots generate five to seven times more hashtags and three times more tweets per peak month than humans, with spikes around the Trump assassination attempt, party conventions, and Biden's withdrawal. Topic modeling shows bots consistently amplifying pro-Trump slogans and media-driven tags (e.g., #foxnews), whereas human discourse is more diverse and fades quickly. Crucially, network analysis reveals that removing bots fragments the reply graph into over six times more communities, demonstrating that bots served as bridges masking underlying fragmentation and artificially inflating cross-group cohesion. These findings indicate that a relatively small bot population can reshape both the content and the topology of political conversation, underscoring the need for robust detection and mitigation before future elections.

Keywords

social bots, bot detection, 2024 U.S. election, Twitter/X, community detection, network centrality, topic modeling, polarization

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1 Introduction

The adoption of social bots has increased exponentially over the past decade, particularly as tools to misinform and mislead the general public in online spaces [9]. Specifically, in high stake events such as presidential elections, the utilization of social bots is greatly amplified [15]. These bots are made to mimic human behavior while also spreading misinformation to sway public opinion at a large scale [29]. Despite their growing presence, social media platforms have not made a significant effort to remove these bots, which allows them to continue influencing public opinion [23]. As social media continues to dominate for political topics, it's vital to understand the full impact and influence that social bots can present.

This study investigates the presence, influence, and impact of social bots on political discourse during the 2024 U.S. presidential election. We examine how bots infiltrate online conversations, amplify polarization, and shape community structures. Our goal is to uncover how automated accounts interact with human users, drive political narratives, and potentially alter public perception.

The 2024 U.S. election presents a timely and underexplored case study. While bots have played roles in past elections, their tactics have evolved, especially with the rise of AI-generated content that makes bot behavior more human-like and harder to detect [7]. At the same time, the sheer scale of social media data and the sophistication of bot networks present new analytical challenges. This paper aims to address those challenges through a multi-stage investigation.

We begin by applying BotMoE, a state-of-the-art bot detection model, to a large-scale dataset of X posts related to the 2024 election [3]. With bot labels in hand, we conduct a series of downstream analyses to examine bot behavior from multiple perspectives: temporal patterns of activity, dominant narratives through hashtag and topic modeling, influence within the social graph, and structural effects on community formation and polarization.

The main contributions of this paper are as follows:

(1) **Large-scale bot detection:** We apply BotMoE, a state-of-the-art bot detection model, to over 42 million election-related tweets, producing one of the largest labeled datasets of the 2024 U.S. election.
(2) **Behavioral comparison:** We analyze temporal patterns and hashtag/topic use, showing that bots sustain high-volume, slogan-driven discourse even as human attention declines.
(3) **Structural impact:** We demonstrate that bots act as *bridges between communities*, masking fragmentation and inflating the appearance of cross-group cohesion.
(4) **Disproportionate influence:** Using multiple centrality metrics, we show that bots occupied disproportionately influential positions, engaging more broadly and effectively than humans.
(5) **Conceptual shift:** We argue that bots' most consequential effect is not only amplifying narratives but also *restructuring the topology of discourse*, altering how political conversations appear to unfold online.

The remainder of this paper is organized as follows: Section 2 provides background on social bots and bot detection techniques. Section 3 reviews related work. Section 4 presents our methodology.

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Section 5 discusses limitations and future work. Section 6 concludes the paper.

2 Background

2.1 Social Bots in Political Discourse

Social bots are automated accounts on social media platforms designed to mimic human behavior while performing high-frequency or large-scale tasks. These tasks can range from benign content dissemination (e.g., news, songs or weather bots) to more malicious activity, such as manipulating political discourse, spreading misinformation and hate, or amplifying partisan narratives. Social bots often blend into human networks by mimicking posting habits, engagement patterns, and language use.

Social bots have played a key role in shaping online political discourse in recent years. For example, during the 2016 U.S. Presidential Election, coordinated bot networks were documented spreading polarizing content, retweeting partisan messages, and even promoting fake news to influence voter sentiment [4]. Similar trends have been observed in other major democratic events, including the Brexit referendum [18] and the 2020 U.S. election [13]. These incidents have raised concerns about the integrity of democratic processes and the effectiveness of platforms at mitigating bot presence.

2.2 Social Network Analysis

Social media platforms naturally form complex networks, where users are nodes and interactions (e.g., follows, replies, retweets) form edges. Analyzing these networks provides insights into how information spreads, how communities are structured, and how influence is distributed across users. Social network analysis is therefore a key tool for understanding bot activity in political contexts.

One relevant concept is *polarization*, which refers to the ideological clustering of users into like-minded groups with limited cross-group interaction. Bots have been shown to exacerbate polarization by preferentially engaging with one side of a political debate and reinforcing echo chambers [28]. *Modularity* [16] is a metric that has been used in research to quantify polarization. It is most commonly used as a community quality metric to measure the level of segregation of two particular groups; such as democrats vs republicans, people in favor or against abortion etc. A network segmentation with high modularity indicates that the social graph can be divided into groups having many internal connections between nodes and few connections to the other group, widely accepted as an indication of polarization [8].

Another useful concept is *centrality*, particularly Degree, PageRank and Eigenvector centrality, which all measure the influence of a node based on its connections to other influential nodes [17]. Bots often attempt to gain central positions in interaction networks to maximize their reach and amplify certain narratives [17]. Studying centrality metrics can reveal whether bots succeed in embedding themselves within influential clusters of political discourse.

Degree centrality specifically measures the influence each node has based on its direct connections with other nodes. PageRank goes a bit deeper and determines the ranking between each node based on the edges, those with more connection and influence get a higher score. Lastly, Eigenvector centrality measures a node's influential behavior and how they influence the other nodes in the network. Using all three of these metrics, we can get a more comprehensive understanding of how each social bot affects the network and their influence.

2.3 Measuring Bot Activity and Behavior

Detecting and characterizing social bots is a fundamental challenge in computational social science. Broadly, bot detection approaches can be categorized into three families: feature-based, text-based and graph-based methods.

Feature-based methods conduct feature engineering based on handcrafted user features derived from metadata and user profile information. Text-based methods use NLP techniques to detect Twitter bots through their tweet content and profile descriptions. Graph-based methods attempt to identify bots based on the graph structure of the user interaction network. They utilize methods such as GNNs.

Modern bots have evolved to escape most popular detection methods, through techniques such as stealing profile information and tweets from authentic users, LLMs and metadata manipulation. Another challenge that faces bot detection methods is the variation in bot behavior and characteristics across user communities in different spaces e.g.: cryptocurrency, politics, product promotions. To combat this challenge, recent bot detection approaches such as BotMoE[22] and LLM-Botguard [10] utilize a Mixture of Experts architecture, which 1) combines all 3 main categories of features to improve robustness towards feature manipulation and 2) learns community-aware features to improve generalization towards the diverse Twitter communities.

3 Related Work

3.1 The Influence of Social Bots on Online Discourse

Social bots are now central to online political discourse, often deployed to amplify candidates, shape narratives, and influence public opinion. Since emerging over 15 years ago, their scale has grown rapidly [9]. During the 2016 U.S. election, bots generated nearly 20% of all Twitter content related to the race [4], underscoring their outsized impact.

Bots do more than increase the volume of political conversation; they strategically amplify partisan narratives, manipulate perceived consensus, and influence the visibility of ideological content. Several studies show that bots actively shape public opinion by amplifying partisan or ideological messaging. During the 2017 U.S. Special Counsel investigation, bots made up 23% of the Twitter accounts but generated a disproportionate 35% of the retweets, largely supporting pro-Trump narratives and simulating a unified far-right community [17]. These bots mimicked legitimate user networks, amplifying fringe views and creating the illusion of widespread consensus to sway public perception. Similarly, research on the Pakistani Twitter space revealed how suspicious accounts coordinated to elevate specific political hashtags and influence visibility [2]. Evidence from Brazil's elections also shows extensive bot coordination, including over 20,000 new Twitter accounts created in a single

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day and accounts managing hundreds of distinct identities[26]. Research also shows that emotions are highly contagious on social media [20], raising concerns that such coordinated bot activity could exploit emotional contagion to amplify polarization and distort public perception on a large scale.

Beyond political amplification, bots play a central role in the spread of hate speech and misinformation. During the COVID-19 pandemic, one study found that while hate speech in online discussions about racism was initially driven by human users, bots soon became the dominant contributors. Over time, the focus of hate speech shifted from racial groups (e.g., "Asian" or "Chinese") to political entities (e.g., "Democrat" or "Republican"), illustrating a bot-driven redirection of discourse in the lead-up to the 2020 U.S. elections [30]. Similarly, bots have been shown to play a key role in the early and rapid dissemination of fake news. These accounts often target influential users, increasing the likelihood that humans will unknowingly amplify false claims. Studies suggest that successful misinformation campaigns are often heavily supported by bots, making their detection and containment a critical priority [29]

Prior work has firmly established the disruptive role of social bots in online discourse, particularly in political contexts. Studies have shown that bots not only amplify partisan narratives but also redirect conversations toward politically charged identities, especially during key events such as elections and public health crises. Their ability to spread misinformation and hate speech at scale, often ahead of widespread human engagement, demonstrates their central role in shaping public opinion. These findings provide a strong foundation for our investigation into the 2024 U.S. presidential elections, motivating a deeper analysis of bot prevalence, temporal activity, influence in interaction networks, and their contribution to polarization and agenda setting.

3.2 Social Bots and the 2024 US Presidential Election

In the 2024 US presidential election, outside actors have shown efforts to use artificial intelligence to drive users towards specific narratives and domains [7]. There are many ways that this can occur: Deepfake images, Robocalls, and social bots to name a few[15]. Social bots have been used in particular to spread disinformation and distrust in election systems and the 2024 US presidential election was no exception [15]. They created visual images and content which 'tricks' the user into believing a story which is not reality but looks convincing to the untrained eye. [6] The full impact to the 2024 election by social bots has not been fully explored but based on the previous information about the 2016 presidential election [4] and some very recent works we know that these actors were present on social media platforms during this last election cycle. This information informs the need to understand how they affect the 2024 presidential election.

3.3 Social Bot Detection Methods

Research on social bot detection has explored how automated accounts operate, their influence and the challenges involved in identifying them. Early studies outlined the transition from basic spam bots to more sophisticated accounts that mimic human behavior and integrate into communities to amplify narratives [14]. A comparative analysis of bot and human characteristics identified key behavioral and structural differences. Bots post more frequently, use more hashtags and mentions, and form denser, star-shaped interaction networks. Humans, by contrast, show more varied identity affiliations, cover a broader range of topics, and engage more often through replies and media sharing [24]. These distinctions point to the evolving complexity of bot behavior and the need for more refined detection methods.

Further analysis showed that newer bots adopt strategies such as occupying the network periphery, targeting niche communities, and imitating influencers to spread content effectively while avoiding detection [27]. The emergence of cyborg accounts, which alternate between human and automated behavior, further complicates detection and demands updated solutions [25]. Experiments have also shown that human users often struggle to distinguish bots from real accounts [19].

To address detection challenges, one approach uses minimal metadata and strategic training data selection to monitor Twitter's public stream in real time with high interpretability and generalization performance [31]). Other approaches combine language models with graph neural networks to enhance performance, particularly against sparse or evasive bots [32]. Techniques such as knowledge distillation transfer information from graph-based models to language models, allowing graphless inference at the test time [5]. Community-aware architectures have also been introduced to better model diverse user behaviors and detect harder-to-spot bots [22]. More recent work addresses large language model-based bots through a mixture-of-experts and graph neural network framework [11].

4 Methodology

4.1 Bot Detection Model Training

We used the BotMoE model to classify each X user account in the 2024 U.S. presidential election dataset as a social bot or a human user. We trained this model on the TwiBot-20[12] bot detection benchmark dataset, which contained over 33 million tweets from 229,573 users and included 11,826 labeled examples of social bots and human users. We followed the original BotMoE's experimental settings[22], including the data-split ratio, hyperparameters, and evaluation metrics, to replicate their model performance. We trained five instances of the model using different random seeds and recorded the mean accuracy and F1 score across these runs. Our experiments achieved a mean accuracy of 87.38% and a mean F1 score of 89.11%, nearly identical to the original BotMoE's reported means of 87.76% and 89.22%, respectively.

4.2 Preprocessing

Using TwiBot-20's preprocessing codebase and libraries, we followed a similar structure to preprocess our code based on the 2024 U.S. presidential election dataset. We began by filtering through the data, formatting similar to TwiBot-20's format and removing duplicates. This resulted in 4,292,568 nodes, our individual users, and 39,168,505 edges, our tweet relations. Next, using TwiBot-20's preprocessing codebase, we adjusted their code to fit our dataset

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and processed our nodes and edges to create PyTorch files, which were later used during our inference for bot detection.

4.3 Temporal Analysis

All tweet timestamps supplied by X are in UTC. We retain that time zone for every analysis. Tweets are aggregated into calendar months for the overview and calendar days for the July 2024 peak. Each tweet inherits the bot or human label assigned by BotMoE. Event markers for Trump assassination attempt, party conventions, Biden's withdrawal, etc. are aligned to their UTC date of occurrence based on news sources. All aggregation, plotting, and topic modeling are performed in Python 3.9 with BERTopic 0.17, Matplotlib 3.9, Plotly 6.2, Pandas 2.3, PyArrow 20, and DuckDB 1.3

4.4 Inference

For inference, the processed files were loaded into the trained BotMoE model to generate bot/human predictions for each user node. To accurately map each model prediction back to the original user account, we maintained a consistent index-to-UID mapping throughout the pipeline. This guaranteed that the final results could be traced back to the correct user IDs for downstream analysis and reporting. Additionally, we combined predictions from our five trained models using a majority-vote strategy to ensure higher confidence in our labels. This resulted in 26% of the users in the network being identified as bots.

4.5 Polarization Analysis

To assess the impact of bots on the polarization of the interaction network, we first constructed a clean version of the network. The initial network contains 4,292,568 nodes and 39,168,505 edges. After excluding non-reply edges and edges that point to nodes that do not exist in our dataset, the number of edges was reduced to 20,259,109.

We used the cugraph library to construct a graph from the edge list. We identified the degree distribution and filter nodes with degree below the first quartile (degree = 4). This dropped the number of nodes to 839,249 and the number of edges to 16,889,302. We then ran the Louvain and Leiden community detection algorithms on the filtered graph. We calculated modularity on the human+bots graph, and on the version of the graph that contains humans only, to see if bots have a significant impact on modularity. To further understand the impact of bots on community structure, we inspected the sizes of communities before and after removing bots, and identified the most influential nodes within the largest communities.

4.6 Centrality Analysis

Using the labeled results from the inference for bot detection, we analyzed the different relationships social bots and human users had in the 2024 U.S. presidential election dataset using NetworkX to graph and calculate metrics and Graphistry to visualize our results.

Our analysis used three key metrics to understand user interactions: PageRank, Degree, and Eigenvector centrality. PageRank estimates how a node influences the network, and comparatively to other nodes in the network. Degree centrality, in all three forms (overall, in, and out) measures the interactions of each node. Lastly, Eigenvector shows how connected the node is in the network, especially to other influential nodes.

In addition, we filtered edges that did not communicate with users in our node set and edges that only contained retweets and replies to focus more on the relationships between users and bots in our node and edge list. After filtering, our dataset contained 12,089,780 edges and 2,582,093 nodes.

5 Results

5.1 Temporal Analysis

We analyzed the temporal distribution of social bot activity in the 2024 U.S. presidential election dataset, to identify patterns and trends in social bot behavior. The results show that social bots exhibit clear patterns of activity, with peaks around key event dates, such as the Trump assassination attempt, the vice presidential debate, and the election day.

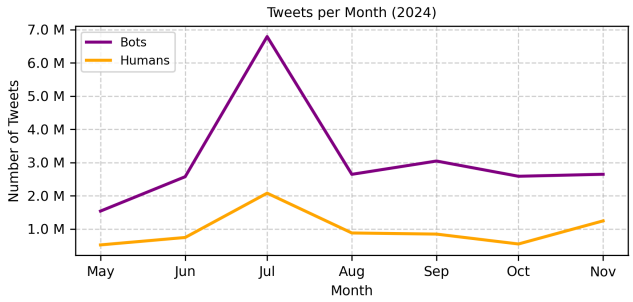


Figure 1: Monthly distribution of tweets by humans and bots in 2024

5.1.1 Temporal Distribution of Tweets. Figure 1 compares the monthly tweet volumes of human users and social bots from May through November 2024. Human activity grows steadily from May to a peak in July, coinciding with the Trump assassination attempt, the Republican National Convention (RNC), and President Biden's withdrawal from the presidential race, before declining in August. Likewise, social bot activity shows a peak in July, followed by a decline in August. This decline reflects a slowdown following the intense activity of July, although activity increased again around Harris's nomination and the Democratic National Convention (DNC) later in the month. The close correlation in activity patterns suggests that social bots are mirroring and amplifying human discussions around pivotal political events.

After preprocessing and running our bot detection inference, the processed data for temporal analysis comprises 1,863,971 human accounts and 1,116,065 bots. Although bot accounts (1,116,065) are outnumbered by human accounts (1,863,971), bots generated a far larger volume of tweets. At its peak in July, bots posted approximately 6.8 million tweets, more than three times the human peak of approximately 2.0 million, and on 14 July alone, bots produced nearly 580,000 tweets compared to approximately 140,000 from humans.

Figure 2 provides a closer look at the daily tweet volumes of human users and social bots in July 2024. The data reveals a significant spike in both human and bot activity around key events,

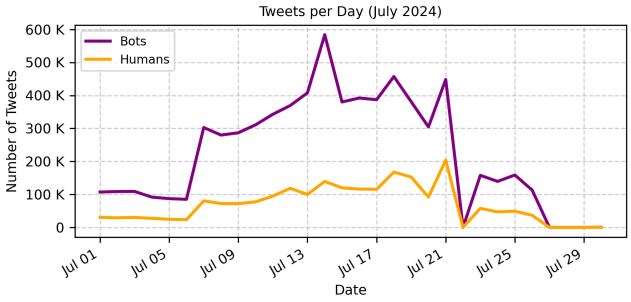


Figure 2: Daily distribution of tweets by humans and bots in July 2024

such as the attempted assassination of Trump on July 13, the Republican National Convention from July 15 to 18, and President Biden's withdrawal announcement on July 21. This indicates that social bots are actively participating in discussions and amplifying the same events as human users.

Finally, moderate peaks appear in September, particularly around September 11 and 14, which correspond to the second presidential debate and the second assassination attempt against Trump, respectively. These smaller September peaks confirm that social bots continued to mirror and amplify human-driven conversations at each major political milestone.

Table 1: Top 20 hashtags used by human users and social bots from May to November 2024

(a) Human users

Tag	Count
trump2024	47,217
maga	34,719
trump	9,530
biden	7,784
bidenharris2024	5,581
donaldtrump	3,223
kamalaharris	2,046
usa	1,639
gop	1,602
maga2024	1,593
election2024	1,572
rnc	1,519
biden2024	1,517
joebiden	1,497
maha	1,495
bitcoin	1,484
news	1,449
nahbabynah	1,378
politics	1,328
jokes	1,222

(b) Social bots

Tag	Count
trump2024	381,049
maga	374,575
trump	165,843
biden	121,148
bidenharris2024	75,382
donaldtrump	44,235
kamalaharris	39,748
gop	38,506
usa	31,273
foxnews	28,900
joebiden	26,653
smartnews	26,651
maha	25,571
election2024	24,963
democrats	22,977
project2025	22,567
trumpvance2024	22,534
tiktok	22,260
americafirst	22,000
biden2024	21,775

5.1.2 Hashtag Usage over Time. We examined the top 20 hashtags used by human users and social bots from May to November 2024 (Table 1). Human users primarily employed political slogans (e.g., *trump2024*, *maga*, *bidenharris2024*), whereas social bots' top tags include both these campaign slogans and a notable presence of media tags (e.g., *foxnews*, *smartnews*, *tiktok*). This pattern suggests bots promote political slogans and drive attention to media platforms.

Although bot accounts (1,116,065) are fewer than human accounts (1,863,971), bots generate a disproportionately high volume of hashtag traffic. Their top hashtag (*trump2024*, 381k uses) appears eight times more often than the human top tag. Despite making up 37.5% of accounts after preprocessing and inference, bots account for eight of the ten most used hashtags.

Figure 3 shows hashtag usage spikes around nearly every major event: the attempted assassination of Trump (July 13), the Republican National Convention (July 15-18), President Biden's withdrawal (July 21), Harris's nomination (Aug 5), the Democratic National Convention (Aug 19-22), the second presidential debate (Sept 11), the second assassination attempt on Trump (Sept 14), the vice presidential debate (Oct 2), Biden's "garbage" comment on Trump supporters (Oct 29), and Election Day (Nov 5). Neutral tags (e.g., *smartnews*, *tiktok*) also surge concurrently with major political events.

Rather than showing a one sided bias, bots mirror both Republican and Democratic slogans (e.g., *trump2024*, *bidenharris2024*) while also driving attention to media brands and trending neutral topics (e.g., *smartnews*, *israel*, *russia*).

5.1.3 Hashtag Topics over Time. To examine how discussion themes shifted over time, we trained separate topic models on hashtags posted by bots and human users. Each model groups related hashtags into a topic. We then tracked the monthly volume of the ten largest topics in each class. The resulting topic trends are plotted in Figures 4 and 5, and the top five keywords defining every topic are listed in tables 2 and 3.

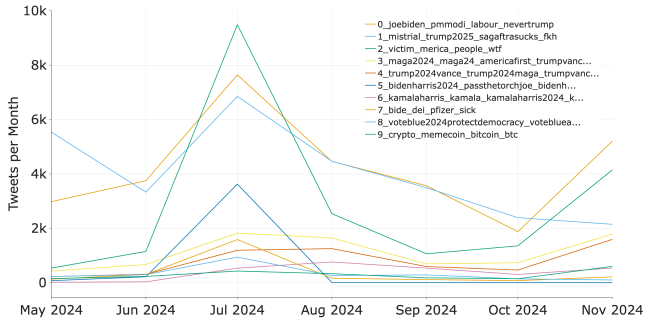
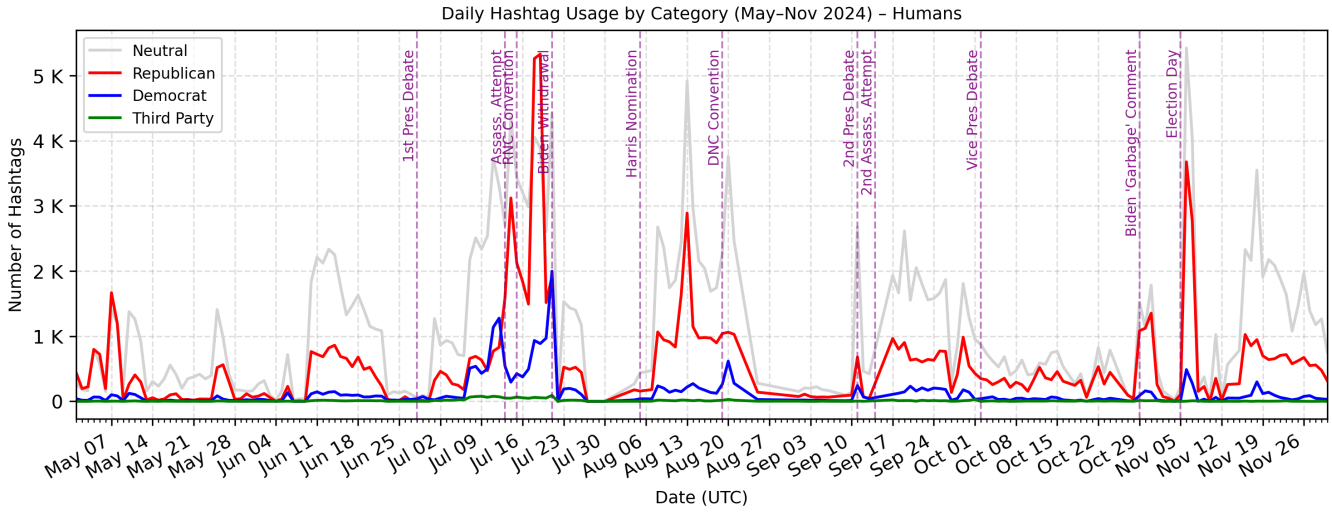
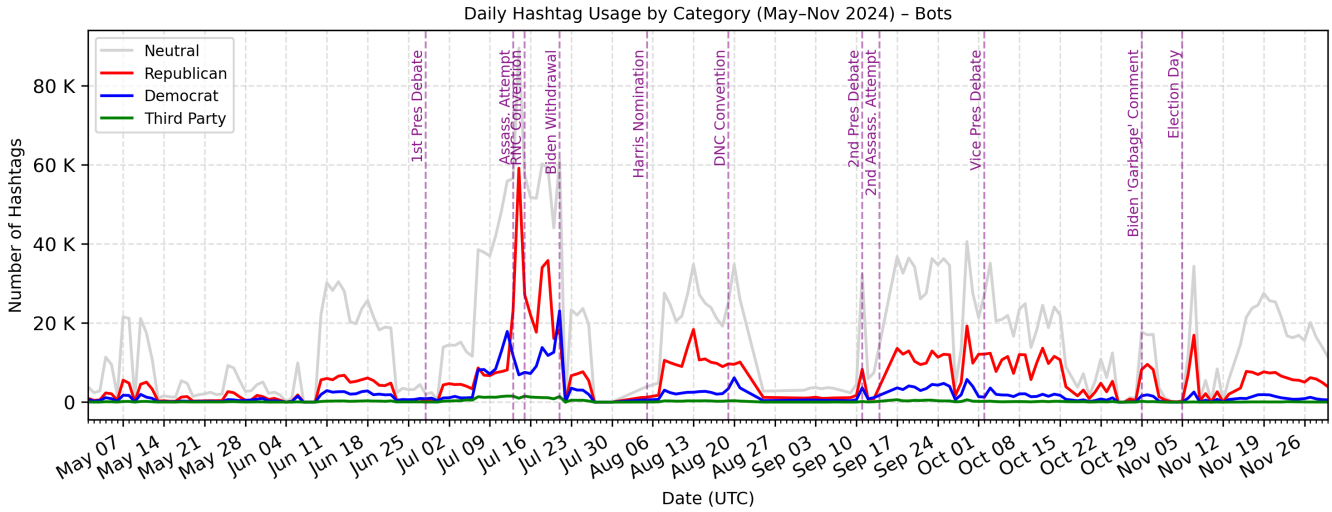


Figure 4: Hashtag topics over time for humans

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(a) Human Accounts



(b) Bot Accounts

Figure 3: Daily hashtag usage by category for human users and social bots in the 2024 U.S. presidential election dataset

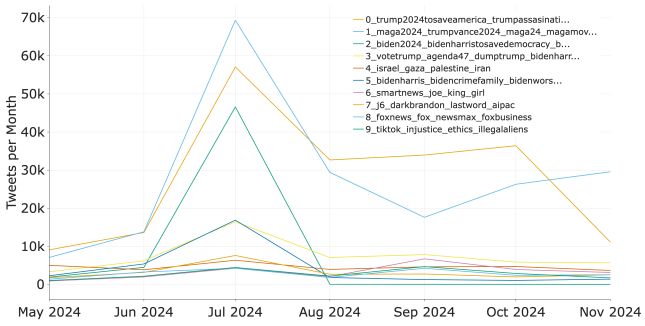


Figure 5: Hashtag topics over time for bots

Table 2: Top hashtag topics by humans

Topic	Count	Representation (Top 5 Terms)
0	29,495	joebiden, pmmodi, labour, nevertrump, conservative
1	28,225	mistrial, trump2025, sagaftrasucks, fkh, 2a
2	20,291	victim, merica, people, wtf, topgun
3	7,793	maga2024, maga24, americafirst, trumpvance2024, magacult
4	5,647	trump2024vance, trump2024maga, trumpvance2024, usaelection2024, election2024
5	3,936	bidenharris2024, passthetorchjoe, bidenharris24, passthetorch, expectus
6	2,737	kamalaharris, kamala, kamalaharris2024, kamala2024, uselection2024
7	2,602	bide, dei, pfizer, sick, google
8	2,274	voteblue2024protectdemocracy, voteblueasifyourlivesdependonit, voteblue, nevervoteconservative, votebluetoprotectwomenrights
9	2,078	crypto, memecoin, bitcoin, btc, solana

Table 3: Top hashtag topics by bots

Topic	Count	Representation (Top 5 Terms)
0	194,011	trump2024tosaveamerica, trumpassasination-attempt, trumpsproject2025, trumpvance2024, trump2024vance
1	193,216	maga2024, trumpvance2024, maga24, magamovement, protrump
2	53,267	biden2024, bidenharristosavedemocracy, bidenharris4moreyears, 4moreyears, finishthejob
3	52,532	votetrump, agenda47, dumptrump, bidenharris4moreyears, donthecon
4	32,296	israel, gaza, palestine, iran, hamas
5	30,217	bidenharris, bidencrimefamily, bidenworstpresidentever, bidenflation, bidenborderinvasion
6	23,122	smartnews, joe, king, girl, damn
7	22,630	j6, darkbrandon, lastword, aipac, bigpharma
8	20,041	foxnews, fox, newsmax, foxbusiness, thefive
9	19,264	tiktok, injustice, ethics, illegalaliens, bordercrisis

Both human users and bots exhibit a pronounced spike in mid to late July, aligned with the Trump assassination attempt, the Republican National Convention, and President Biden's withdrawal, as shown in topics 0-3 in both models. These topics show that in July 2024, human users were more supportive of Biden and more anti-Trump as indicated by key terms such as *joebiden*, *nevertrump*, and *magacult*. In contrast, bots heavily amplified pro-Trump narratives, using terms such as *trump2024tosaveamerica*, *protrump*, and *votetrump*.

After the intense activity during July, human hashtag volume slows down until a modest uptick in November (topics 0-2), while topics 3 and 4 linked to Trump campaign events consistently hover near 1,000 tags/month. In contrast, bot hashtag activity also declines after July, yet its pro-Trump and MAGA clusters (topics 0 and 1) remain roughly five to seven times higher than the corresponding human volumes from August through October. This mirrors our earlier observation that bots maintain higher tweet volumes even when human interest decreases (Sec. 5.1.1).

Bots concentrate on two themes: (i) pro-Trump and MAGA slogans (topics 0 and 1) and (ii) media amplification tags that drive traffic to outlets such as *FoxNews*, *Newsmax*, *SmartNews*, or *TikTok* (topics 6, 8, and 9). In contrast, human users show a broader mix of interest. Their largest clusters cover pro-Biden (topic 0), right-wing chatter (e.g. *mistrial*, *trump2025*, and *sagaftrasucks*) (topic 1), and reactions such as *victim*, *people*, and *wtf*. The remaining human topics are varied (e.g. Harris, cryptocurrency) but small, each under roughly 1,000/month.

In short, bots not only sustain far greater hashtag volumes and concentrate that activity on partisan slogans and media-driven narratives, whereas human discourse is more diverse and fades more quickly because of its much lower hashtag traffic.

5.2 Polarization Analysis

To assess the role of bots in shaping community structure and potential polarization, we performed community detection before and after bot removal. Table 4 summarizes the results using Louvain and Leiden algorithms.

Interestingly, while modularity remains relatively stable, indicating that the overall level of polarization in the network does not drastically shift, the number of detected communities increases sharply after removing bots. For example, the Louvain algorithm detects a sixfold increase in communities (from 5,580 to 32,723), and Leiden detects nearly a sixfold rise as well (from 82 to 481). This dramatic fragmentation suggests that bots were not merely passive participants but actively served as structural bridges, linking otherwise separate clusters. Bots may have artificially connected users from different communities, giving the illusion of greater cohesion or dialogue across sub-groups. Once removed, the underlying fragmented landscape becomes more apparent.

Figure 6 further illustrates this phenomenon by comparing the distribution of community sizes. Before bot removal, two large communities dominated the network—comprising 89% of nodes in Louvain and 70% in Leiden. After bot removal, these large communities splinter into many smaller ones, revealing a more fragmented topology.

To gain insight into the composition of the largest communities, we examine the most influential nodes—based on the PageRank metric—within the two largest communities identified by the Louvain algorithm. In the first community, the top-ranking accounts are @DemocraticWins (known as @KamalasWins at the time of data collection) and @JoeBiden, reflecting a politically left-leaning cluster. In contrast, the second community is dominated by @ElonMusk and @catturd2, the latter being a user with 3.8 million followers, described by Wikipedia as "a right-wing American Twitter shitposter and Internet troll, Phillip Buchanan" [1]. This contrast between the left-leaning and right-leaning dominant users underscores an ideological divide within the largest communities. Notably, when bots are removed, these polarized communities fragment into numerous smaller groups. This suggests that bots play a significant role in maintaining the appearance of large, cohesive groups by amplifying connections among politically aligned users, reinforcing polarization and community structure as also observed in prior work [17]

Algorithm	Metric	Before Bots Removed	After Bots Removed
Louvain	Modularity	0.532	0.538
Louvain	Number of Communities	5,580	32,723
Leiden	Modularity	0.3452	0.3339
Leiden	Number of Communities	82	481

Table 4: Community detection metrics using Louvain and Leiden algorithms before and after removing bots.

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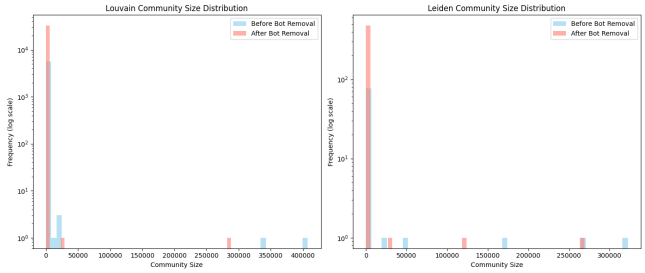


Figure 6: Community size distributions before and after removing bots for Louvain (left) and Leiden (right).

5.3 Centrality Analysis

From our labeled users and focused edges, we conducted a centrality analysis on 12,089,780 edges which contained 2,582,093 users. On average, each bot had 3.68 interactions or targets towards human users. Notably, some users are targeted by bots on a larger scale than others, particularly those who are prominent figures in political and social worlds. This could indicate a social bot strategy, where they are targeting major figures so that their influence could be greater.

For our analysis, we conduced several different testing measures to find the centrality of bots in our network. We used PageRank, degree (overall, in-degree and out-degree), and eigenvector to measure this. PageRank results showed that social bots were 228% more influential than human users, showing that bots not only are central in our network, but they are targeting and communicating with other major nodes. Overall degree analysis displayed that bots were 313% more influential than human users, while in-degree calculations, bots were 565% more interacted with than other users. Furthermore, out-degree calculations showed that bots displayed 165% more interactions with other users than humans. Lastly, the eigenvector centrality metric showed that bots were 778% more involved in the general network than human users. These results show how connected the bots are in the network compared to human users.

6 Conclusion

Our analysis reveals that social bots played a significant and multifaceted role in shaping online political discourse during the 2024 U.S. presidential election. Temporally, bots mirrored and amplified human activity, with tweet and hashtag volumes peaking around key political events. Despite being fewer in number, bots generated far more content than human users, particularly during pivotal moments such as the Trump assassination attempt and the party conventions. Hashtag and topic analyses show that bots consistently pushed partisan slogans, especially pro-Trump narratives, and drove attention to media-related content, maintaining high activity levels even as human engagement declined.

Community and centrality analyses further show the structural impact of bots. While overall modularity remained stable, bot removal revealed a dramatic increase in the number of communities, exposing an underlying fragmentation previously masked by bot-driven connectivity. Bots acted as bridges between otherwise disconnected groups, artificially inflating the appearance of cohesion. Centrality metrics confirm that bots occupied highly influential positions within the network—targeting key users, engaging broadly, and amplifying content at significantly higher rates than humans.

Together, these findings indicate that bots are not merely background noise but active agents shaping the structure, content, and perceived polarization of political conversations online.

Our findings highlight the critical need to better understand social bot activity in political discourse, develop more advanced methods for detecting sophisticated bots, and assess their potential influence on public opinion during elections. We aim for this work to support ongoing efforts to analyze and mitigate the effects of social bots on social media, particularly during high-stakes events such as national elections.

7 Limitations and Future Work

The limitations of our work primarily stem from the reliance on existing datasets, which may not fully capture the dynamic nature of advanced social bots and their sophisticated behaviors during the 2024 U.S. presidential election. The benchmark datasets we are using, such as TwiBot-20 or TwiBot-22, may not help us detect the latest social bot techniques with LLMs, which have become increasingly prevalent on social media platforms. These datasets were developed before the widespread use of LLMs in social bots. As a result, our BotMoE model may not detect the nuances of these advanced bots effectively, as they may exhibit human-like behaviors that are not represented in the training data.

Detecting LLM-driven social bots on social media platforms like X poses a significant challenge yet remains underexplored in the existing literature. For future work, we plan to address these limitations by exploring more recent datasets that include social bots utilizing LLMs. For instance, the Masquerade-23 dataset [21] contains tweet-like posts written by LLM-driven social bots on the synthetic Chirper.ai network. By incorporating such datasets, we can develop new methods to detect these advanced bots and improve the BotMoE model's performance on detecting advanced social bots. For example, LLM-BotGuard [11] is a recent work that leverages the Masquerade-23 dataset and proposes a novel model to detect LLM-driven social bots by capturing the unique characteristics of these advanced bots. We can explore similar approaches to enhance our BotMoE model's capabilities.

Additionally, in our network analysis (particularly in centrality and polarization) we currently rely solely on reply edges, which may limit our understanding of broader interaction dynamics. Future extensions should incorporate multiple edge types such as retweets, quotes, and mentions to build a more comprehensive network structure. Finally, while we use BotMoE for labeling accounts, relying on a single model may introduce bias or blind spots. We plan to incorporate multiple bot detection models in the future to increase labeling robustness and confidence.

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